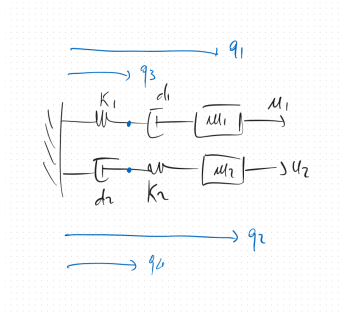




$$\begin{aligned}
 T &= \frac{1}{2} m_1 \dot{q}_1^2 + \frac{1}{2} m_2 \dot{q}_2^2 \\
 U &= \frac{1}{2} k_1 q_3^2 + \frac{1}{2} k_2 (q_2 - q_4)^2 \\
 F &= \frac{1}{2} d_1 (\dot{q}_1 - \dot{q}_3)^2 + \frac{1}{2} d_2 \dot{q}_4^2 \\
 L &= T - U \\
 &= \frac{1}{2} m_1 \dot{q}_1^2 + \frac{1}{2} m_2 \dot{q}_2^2 - \frac{1}{2} k_1 q_3^2 - \frac{1}{2} k_2 (q_2 - q_4)^2
 \end{aligned}$$

	$\frac{\partial L}{\partial \dot{q}}$	$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}}$	$\frac{\partial L}{\partial q}$	$\frac{\partial F}{\partial \dot{q}}$	u	E-L
$i=1$	$m_1 \dot{q}_1$	$m_1 \ddot{q}_1$	0	$d_1 (\dot{q}_1 - \dot{q}_3)$	u_1	$m_1 \ddot{q}_1 - 0 + d_1 (\dot{q}_1 - \dot{q}_3) = u_1$
$i=2$	$m_2 \dot{q}_2$	$m_2 \ddot{q}_2$	$-k_2 (q_2 - q_4)$	0	u_2	$m_2 \ddot{q}_2 - (-k_2 (q_2 - q_4)) + 0 = u_2$
$i=3$	0	0	$-k_1 q_3$	$-d_1 (\dot{q}_1 - \dot{q}_3)$	0	$-(-k_1 q_3) - d_1 (\dot{q}_1 - \dot{q}_3) = 0$
$i=4$	0	0	$k_2 (q_2 - q_4)$	$d_2 \dot{q}_4$	0	$-k_2 (q_2 - q_4) + d_2 \dot{q}_4 = 0$

$$\begin{aligned}
 m_1 \ddot{q}_1 + d_1 (\dot{q}_1 - \dot{q}_3) &= u_1 \\
 m_2 \ddot{q}_2 + k_2 (q_2 - q_4) &= u_2 \\
 k_1 q_3 - d_1 (\dot{q}_1 - \dot{q}_3) &= 0 \\
 -k_2 (q_2 - q_4) + d_2 \dot{q}_4 &= 0
 \end{aligned}$$

$$\begin{aligned}
 d_2 \dot{q}_4 &= k_2 (q_2 - q_4) \\
 \dot{q}_4 &= \frac{k_2}{d_2} (q_2 - q_4)
 \end{aligned}$$

$$\begin{aligned}
 k_1 q_3 - d_1 (\dot{q}_1 - \dot{q}_3) &= 0 \\
 d_1 (\dot{q}_1 - \dot{q}_3) &= k_1 q_3 \\
 \dot{q}_1 - \dot{q}_3 &= \frac{k_1}{d_1} q_3 \\
 \dot{q}_3 &= \dot{q}_1 - \frac{k_1}{d_1} q_3
 \end{aligned}$$

$$\begin{aligned}
 m_2 \ddot{q}_2 + k_2 (q_2 - q_4) &= u_2 \\
 \ddot{q}_2 &= -\frac{k_2}{m_2} (q_2 - q_4) + \frac{1}{m_2} u_2
 \end{aligned}$$

$$\begin{aligned}
 m_1 \ddot{q}_1 + d_1 (\dot{q}_1 - \dot{q}_3) &= u_1 \\
 \ddot{q}_1 &= -\frac{d_1}{m_1} (\dot{q}_1 - \dot{q}_3) + \frac{1}{m_1} u_1
 \end{aligned}$$

$$x_1 = q_1$$

$$\begin{aligned}
x_2 &= \dot{q}_1 \\
x_3 &= q_2 \\
x_4 &= \dot{q}_2 \\
x_5 &= q_3 \\
x_6 &= q_4
\end{aligned}$$

$$\begin{aligned}
\dot{x}_6 &= \dot{q}_4 \\
&= \frac{k_2}{d_2}(q_2 - q_4) \\
&= \frac{k_2}{d_2}(x_3 - x_6)
\end{aligned}$$

$$\begin{aligned}
\dot{x}_5 &= \dot{q}_3 \\
&= \dot{q}_1 - \frac{k_1}{d_1} q_3 \\
&= x_2 - \frac{k_1}{d_1} x_5
\end{aligned}$$

$$\begin{aligned}
\dot{x}_4 &= \ddot{q}_2 \\
&= -\frac{k_2}{m_2}(q_2 - q_4) + \frac{1}{m_2} u_2 \\
&= -\frac{k_2}{m_2}(x_3 - x_6) + \frac{1}{m_2} u_2
\end{aligned}$$

$$\begin{aligned}
\dot{x}_3 &= \dot{q}_2 \\
&= x_4
\end{aligned}$$

$$\begin{aligned}
\dot{x}_2 &= \ddot{q}_1 \\
&= -\frac{d_1}{m_1} \left(x_2 - \left(x_2 - \frac{k_1}{d_1} x_5 \right) \right) + \frac{1}{m_1} u_1 \\
&= -\left(\frac{k_1}{m_1} x_5 \right) + \frac{1}{m_1} u_1
\end{aligned}$$

$$\begin{aligned}
\dot{x}_1 &= \dot{q}_1 \\
&= x_2
\end{aligned}$$

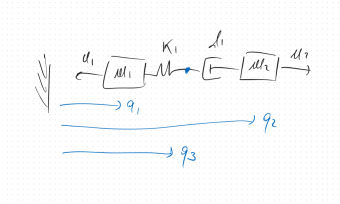
$$\begin{aligned}
A &= \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{k_1}{m_1} & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{k_2}{m_2} & 0 & 0 & \frac{k_2}{m_2} \\ 0 & 1 & 0 & 0 & -\frac{k_1}{d_1} & 0 \\ 0 & 0 & \frac{k_2}{d_2} & 0 & 0 & -\frac{k_2}{d_2} \end{pmatrix} \\
B &= \begin{pmatrix} 0 & 0 \\ \frac{1}{m_1} & 0 \\ 0 & 0 \\ 0 & \frac{1}{m_2} \\ 0 & 0 \\ 0 & 0 \end{pmatrix}
\end{aligned}$$

$y(t)$ sono le posizioni delle masse

$$y(t) = \begin{pmatrix} q_1(t) \\ q_2(t) \end{pmatrix}$$

$$\begin{aligned}
&= \begin{pmatrix} x_1(t) \\ x_3(t) \end{pmatrix} \\
C &= \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix} \\
D &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
y(t) &\text{ è la forza totale applicata a } m_2 \\
&= u_2 - k_2 (q_2 - q_4) \\
&= u_2 - k_2 x_3 + k_2 x_6 \\
C &= \begin{pmatrix} 0 & 0 & -k_2 & 0 & 0 & k_2 \end{pmatrix} \\
D &= \begin{pmatrix} 0 & 1 \end{pmatrix}
\end{aligned}$$



$$\begin{aligned}
T &= \frac{1}{2} m_1 \dot{q}_1^2 + \frac{1}{2} m_2 \dot{q}_2^2 \\
U &= \frac{1}{2} k_1 (q_3 - q_1)^2 \\
F &= \frac{1}{2} d_1 (\dot{q}_2 - \dot{q}_3)^2 \\
L &= \frac{1}{2} m_1 \dot{q}_1^2 + \frac{1}{2} m_2 \dot{q}_2^2 - \frac{1}{2} k_1 (q_3 - q_1)^2
\end{aligned}$$

	$\frac{\partial L}{\partial \dot{q}}$	$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}}$	$\frac{\partial L}{\partial q}$	$\frac{\partial F}{\partial \dot{q}}$	u	E-L
$i=1$	$m_1 \dot{q}_1$	$m_1 \ddot{q}_1$	$k_1 (q_3 - q_1)$	0	$-u_1$	$m_1 \ddot{q}_1 - k_1 (q_3 - q_1) = -u_1$
$i=2$	$m_2 \dot{q}_2$	$m_2 \ddot{q}_2$	0	$d_1 (\dot{q}_2 - \dot{q}_3)$	u_2	$m_2 \ddot{q}_2 + d_1 (\dot{q}_2 - \dot{q}_3) = u_2$
$i=3$	0	0	$-k_1 (q_3 - q_1)$	$-d_1 (\dot{q}_2 - \dot{q}_3)$	0	$k_1 (q_3 - q_1) - d_1 (\dot{q}_2 - \dot{q}_3) = 0$

$$\begin{aligned}
m_1 \ddot{q}_1 - k_1 (q_3 - q_1) &= -u_1 \\
m_2 \ddot{q}_2 + d_1 (\dot{q}_2 - \dot{q}_3) &= u_2 \\
k_1 (q_3 - q_1) - d_1 (\dot{q}_2 - \dot{q}_3) &= 0
\end{aligned}$$

$$\begin{aligned}
d_1 (\dot{q}_2 - \dot{q}_3) &= k_1 (q_3 - q_1) \\
\dot{q}_2 - \dot{q}_3 &= \frac{k_1}{d_1} (q_3 - q_1) \\
\dot{q}_3 &= \dot{q}_2 - \frac{k_1}{d_1} (q_3 - q_1)
\end{aligned}$$

$$\begin{aligned}
m_2 \ddot{q}_2 &= u_2 - d_1 (\dot{q}_2 - \dot{q}_3) \\
\ddot{q}_2 &= \frac{1}{m_2} u_2 - \frac{d_1}{m_2} (\dot{q}_2 - \dot{q}_3)
\end{aligned}$$

$$m_1 \ddot{q}_1 - k_1 (q_3 - q_1) = -u_1$$

$$\begin{aligned}
m_1 \ddot{q}_1 &= -u_1 + k_1 (q_3 - q_1) \\
\ddot{q}_1 &= -\frac{1}{m_1} u_1 + \frac{k_1}{m_1} (q_3 - q_1)
\end{aligned}$$

$$\begin{aligned}
x_1 &= q_1 \\
x_2 &= \dot{q}_1 \\
x_3 &= q_2 \\
x_4 &= \dot{q}_2 \\
x_5 &= q_3
\end{aligned}$$

$$\begin{aligned}
\dot{x}_5 &= \dot{q}_3 \\
&= x_4 - \frac{k_1}{d_1} (x_5 - x_1) \\
\dot{x}_4 &= \ddot{q}_2 \\
&= \frac{1}{m_2} u_2 - \frac{d_1}{m_2} (\dot{q}_2 - \dot{q}_3) \\
&= \frac{1}{m_2} u_2 - \frac{d_1}{m_2} \left(x_4 - x_4 - \frac{k_1}{d_1} (x_5 - x_1) \right) \\
&= \frac{1}{m_2} u_2 + \frac{k_1}{m_2} x_5 - \frac{k_1}{m_2} x_1 \\
\dot{x}_3 &= \dot{q}_2 \\
&= x_4 \\
\dot{x}_2 &= \ddot{q}_1 \\
&= -\frac{1}{m_1} u_1 + \frac{k_1}{m_1} (x_5 - x_1) \\
\dot{x}_1 &= \dot{q}_1 \\
&= x_2
\end{aligned}$$

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ -\frac{k_1}{m_1} & 0 & 0 & \frac{k_1}{m_1} & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -\frac{k_1}{m_2} & 0 & 0 & \frac{k_1}{m_2} & 0 \\ \frac{k_1}{d_1} & 0 & 0 & 1 & -\frac{k_1}{d_1} \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & 0 \\ -\frac{1}{m_1} & 0 \\ 0 & 0 \\ 0 & \frac{1}{m_2} \\ 0 & 0 \end{pmatrix}$$

$$\begin{aligned}
y(t) &= q_1(t) \\
&= x_1(t) \\
C &= (1 \ 0 \ 0 \ 0 \ 0) \\
D &= 0
\end{aligned}$$

Esempio 1. Il pagamento del parcheggio può essere effettuato solo con monete da 1€ e 2€. Con quante sequenze possibili si possono pagare t €?

$x(t)$ è il numero di sequenze con cui posso pagare $t \in \mathbb{N}$

$$t=1, (1), x(1)=1$$

$$t=2, (11, 2), x(2)=2$$

$$t=3, (111, 21, 12), x(3)=3$$

$$t=4, (1111, 211, 121, 112, 22), x(4)=5$$

$$\begin{aligned} x(t) &= \underbrace{x(t-1)}_{\text{modi per pagare } (t-1)\epsilon+1} + \underbrace{x(t-2)}_{\text{modi per pagare } (t-2)\epsilon+2} \\ t &\leftarrow t+2 \\ x(t+2) &= x(t+1) + x(t) \end{aligned}$$

$$\begin{aligned} x_1(t) &= x(t) \\ x_2(t) &= x(t+1) \end{aligned}$$

$$\begin{aligned} x_1(t+1) &= x(t+1) \\ &= x_2(t) \\ x_2(t+1) &= x(t+2) \\ &= x(t+1) + x(t) \\ &= x_2(t) + x_1(t) \end{aligned}$$

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\begin{aligned} \begin{pmatrix} x_1(1) \\ x_2(1) \end{pmatrix} &= \begin{pmatrix} x(1) \\ x(2) \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ 2 \end{pmatrix} \end{aligned}$$